

CSE 4615:: Wireless Networks

Understanding Medium Access Control Mechanisms For Managing Shared Wireless Channels

Lecture 5

Course Teacher:

Ashrafal Alam Khan
Assistant Professor, CSE, IUT

Content Contributors:

Kyle Jamieson, Princeton University

Today

- 1. How does channel access differ between wireless and wired networks?**
2. Medium access control: Provisioned protocols
3. Contention-based medium access control

Wireless is less reliable

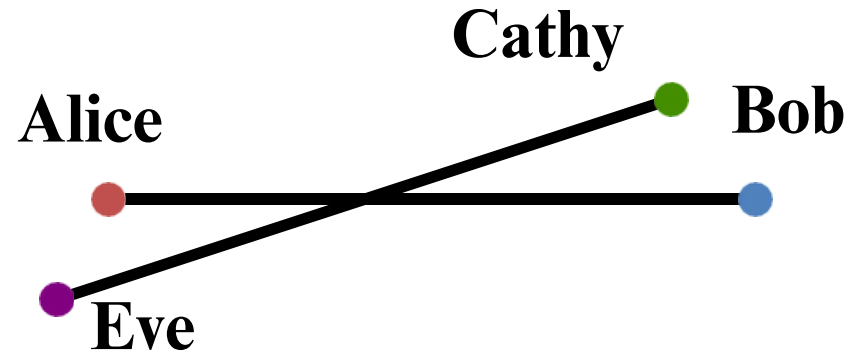


- In wired networks, link **bit error rate** is **10^{-12} and less**
- Wireless networks are **far from that target**
 - Bit error rates of **10^{-6} and above** are common!
- *Why?*

Wireless is a shared medium

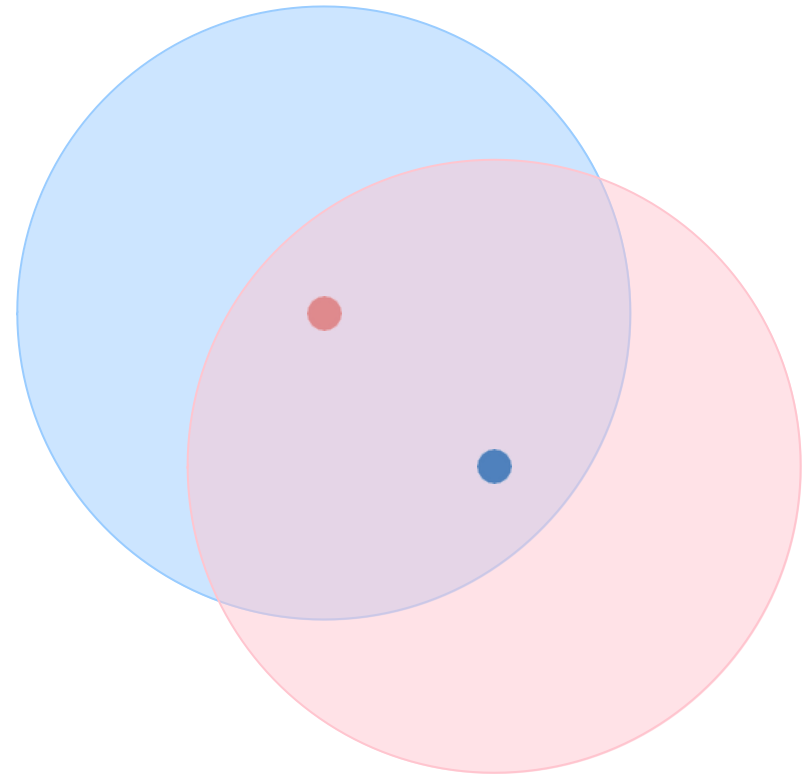
- **Wired networks:**

Alice and Bob's conversation is **independent of** Cathy and Eve's conversation



- **Wireless networks:**

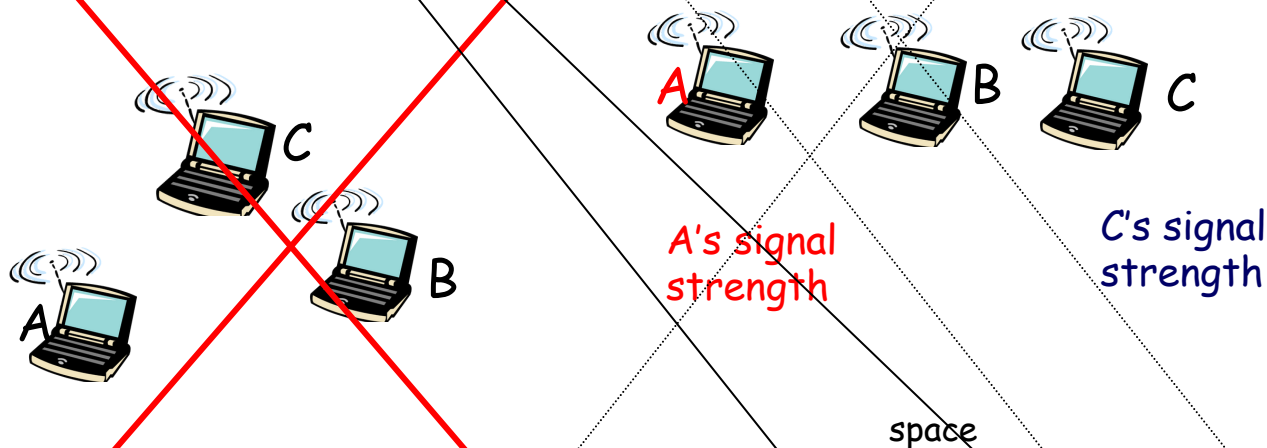
Close by **wireless** conversations **share the same wireless medium**



CSMA: Carrier Sense, Multiple Access



- Multiple access: channel is shared medium
 - Station: wireless host or access point
 - Multiple stations may want to transmit at same time
- Carrier sense: sense channel before sending
 - Station doesn't send when channel is busy
 - To prevent collisions with ongoing transfers
 - But, detecting ongoing transfers isn't always possible



Collision Avoidance & Collision Detection



- Collision detection in wired Ethernet
 - Station listens (receives) while transmitting
 - Detects collision with other transmission
 - Aborts transmission and tries sending again
- Problem #1 in Wireless: cannot detect all collisions
 - Hidden terminal problem
 - Fading
- Problem #2 in Wireless: listening while sending
 - Strength of received signal is much smaller
 - Expensive to build hardware that detects collisions
- So, 802.11 does *not* do collision detection

Medium Access Control in 802.11



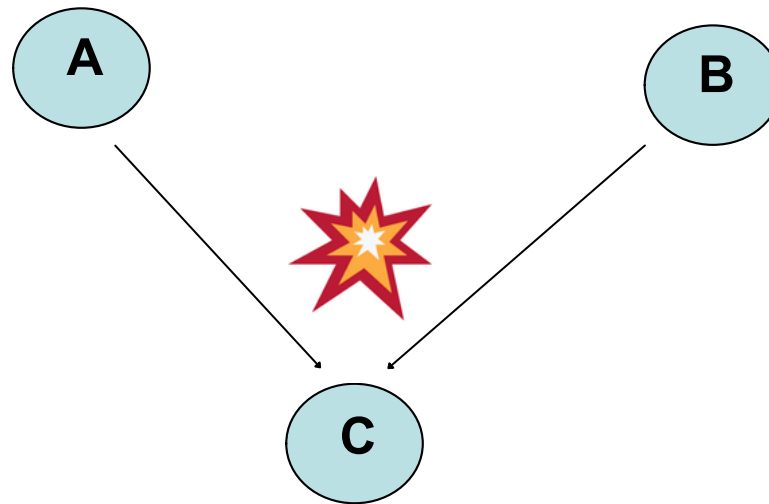
- Collision avoidance, not detection
 - Once a station starts transmitting, send in its entirety
 - More aggressive collision-avoidance techniques
 - E.g., waiting a little after sensing an idle channel
 - To reduce likelihood two stations transmit at once
- Link-layer acknowledgment and retransmission
 - CRC to detect errors
 - Receiving station sends an acknowledgment
 - Sending station retransmits if no ACK is received
 - Giving up after a few failed transmissions

Channel Access through CA: Collision Avoidance, Not Detection

- Listen to medium and wait until it is free (no one else is talking)
- Wait a random backoff time
- **Advantage:** Simple to implement
- **Disadvantage:** Cannot recover from a collision

Wireless Interference

- Two transmitting stations interfere with each other at the receiver
- Receiver gets garbage



Collision Detection and CSMA-CD

- **Procedure**

- Listen to medium and wait until it is free
- Start talking, but listen to see if someone else starts talking too
- If collision, stop; start talking after a random backoff time

- Used for hub-based Ethernet

- **Advantage:** More efficient than basic CSMA

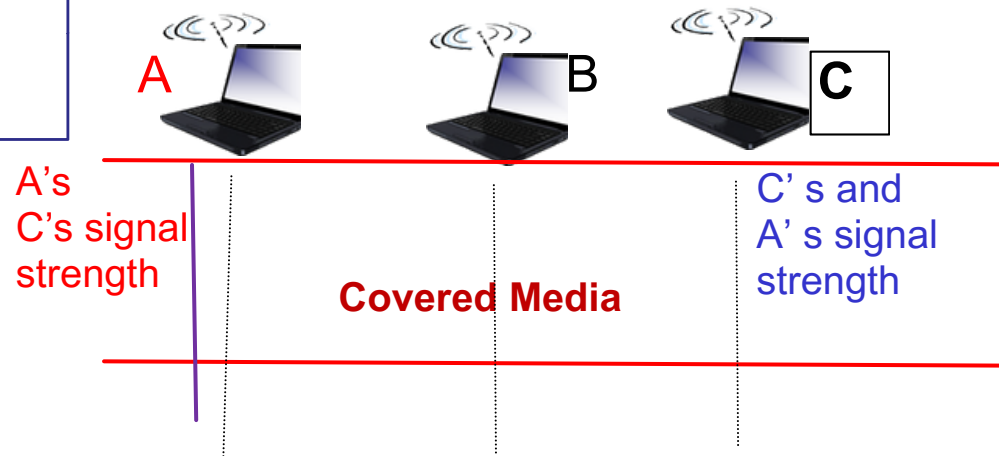
- **Disadvantage:** Requires ability to detect collisions by transmitting station.

- Require additional hardware antenna for concurrent receiving while transmitting (Radios are full duplex)
- More difficult in wireless scenario due to significant signal attenuation

Collision Detection in Wired Network

Here, **Signal State** at Receiver (node B) and Transmitter (Node A) are **almost Identical**.
‘Fate sharing’ of the link’

Collision experienced at Receiver (node B) while both node A and C are transmitting



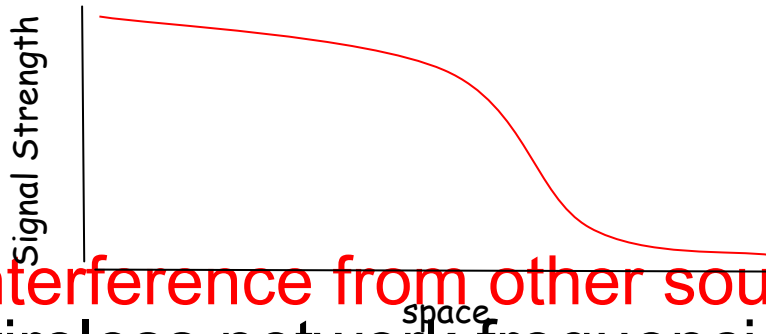
Hence, **Collision Detection at Transmitter (Node A) is Feasible**, by **sensing**(receiving) the **signal strength available now** by **additional receiving antenna while transmission is going on**.

Wireless Link Characteristics

Wireless is a broadcast medium!

Differences from wired link

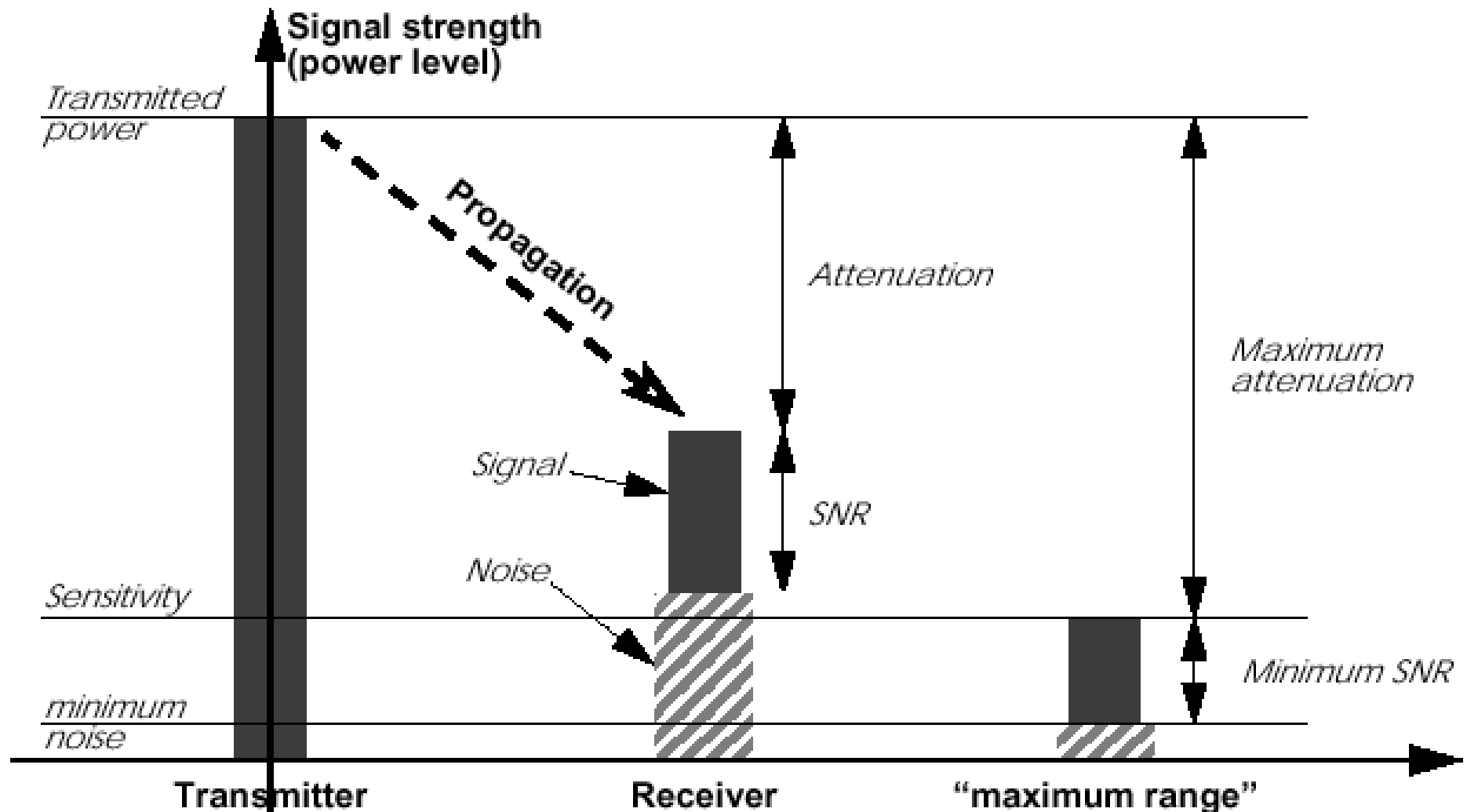
- **decreased signal strength:** radio signal attenuates as it propagates through matter (path loss)



- **interference from other sources:** standardized wireless network frequencies (e.g., 2.4 GHz) shared by other devices (e.g., phone); devices (motors, microwaves) interfere as well

.... make communication across (even a point to point) wireless link much more "difficult"

Wireless Link Characteristics: Attenuation & Propagation range

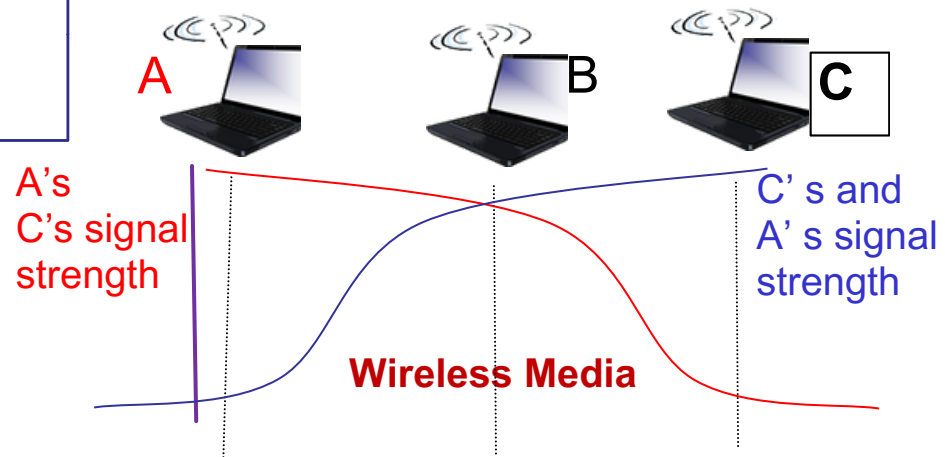


Collision Detection in Wireless Network

Here, **Signal State** at Receiver (node B) and Transmitter (Node A) are **NOT Identical**.

‘No Fate sharing’ of the link’

Collision experienced at Receiver (node B) while both node A and C are transmitting



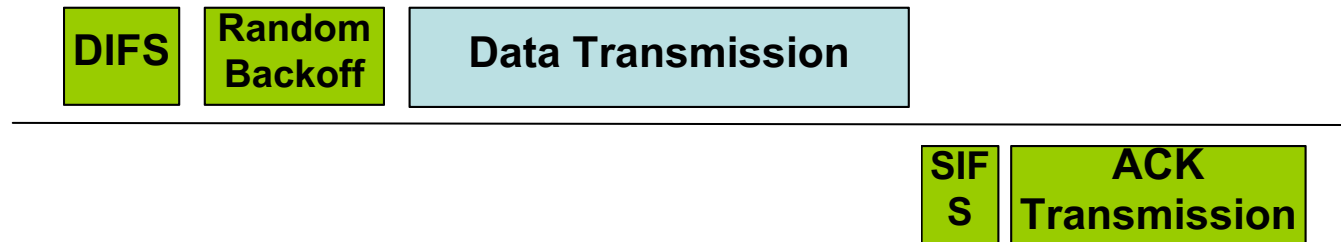
Hence, **Collision Detection at Transmitter (Node A) is NOT Feasible**, by **sensing**(receiving) the **signal strength available now** by **additional receiving antenna while transmission is going on**.

Failure to Detect Collision in Wireless

- No “fate sharing” of the link
 - High loss rates
 - Variable channel conditions
- Transmit signal is stronger than received signal
- Hence, Cannot take advantage of detecting multiple transmission by simultaneously transmit and receive
- Hence, Radios are not full duplex

Solution: Link-Layer Acknowledgments

- Absence of ACK from receiver signals packet loss to sender
- Sender interprets packet loss as being caused by collision



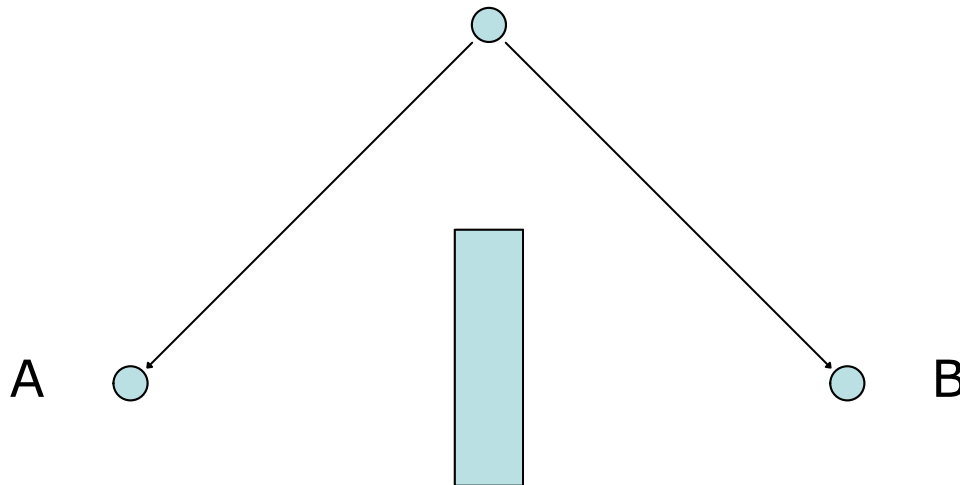
Problem: Does not handle hidden terminal cases.

Carrier Sense Multiple Access with Collision *Avoidance* (CSMA-CA)

- Similar to CSMA but *control* frames are exchanged instead of data packets
 - **RTS**: request to send
 - **CTS**: clear to send
 - **DATA**: actual packet
 - **ACK**: acknowledgement

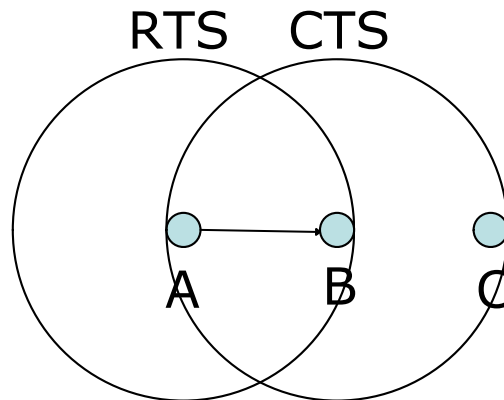
Carrier Sense Multiple Access with Collision *Avoidance* (CSMA-CA)

- Small control frames lessen the cost of collisions (when data is large)
- RTS + CTS provide “virtual carrier sense”
 - protects against hidden terminal



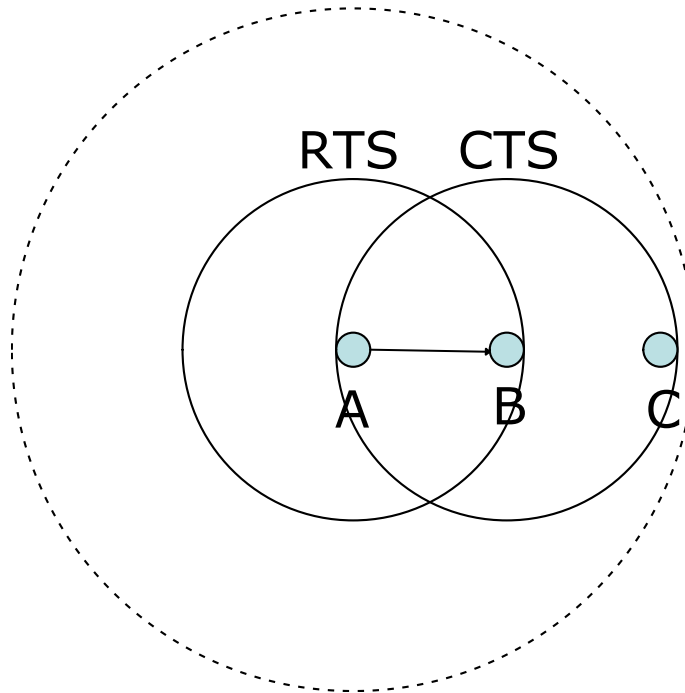
Virtual Carrier Sense

- Provided by RTS & CTS
- Prevents hidden terminal collisions



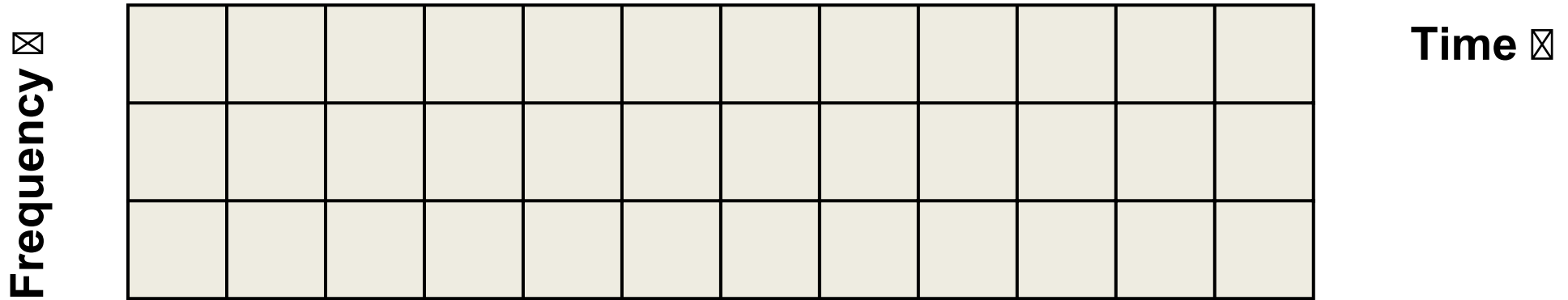
Hidden Terminal Revisited

- RTS transmission by **node A**, retrains all the **neighbors of A** from transmission during A->B transmission.
- RTS transmission by **node B**, retrains all the **neighbors of B** from transmission during A->B transmission.



Understanding Medium Access: The Problem

Medium access: The Problem



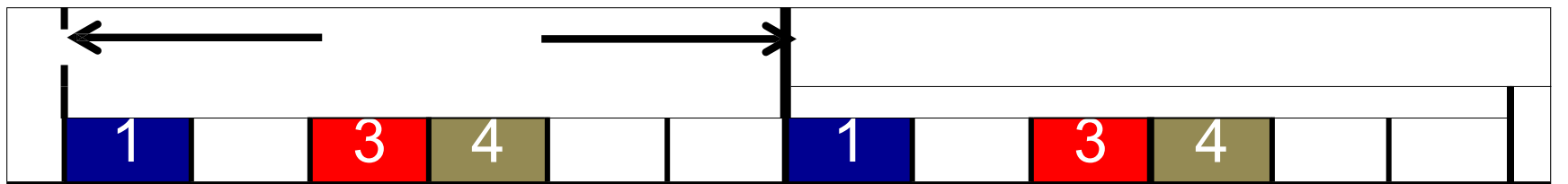
- Receiver must hear ≤ 1 strong transmission at a time, else **collision** (transmissions are lost)
- Two questions:
 1. **How** should the shared medium be **divided**?
 2. **Who** gets to use each such unit of division, and **when**?
- A **medium access control (MAC) protocol** specifies the above

Today

1. How does channel access differ between wireless and wired networks?
- 2. Medium access control: Provisioned protocols**
3. Contention-based medium access control

TDMA: Time Division Multiple Access

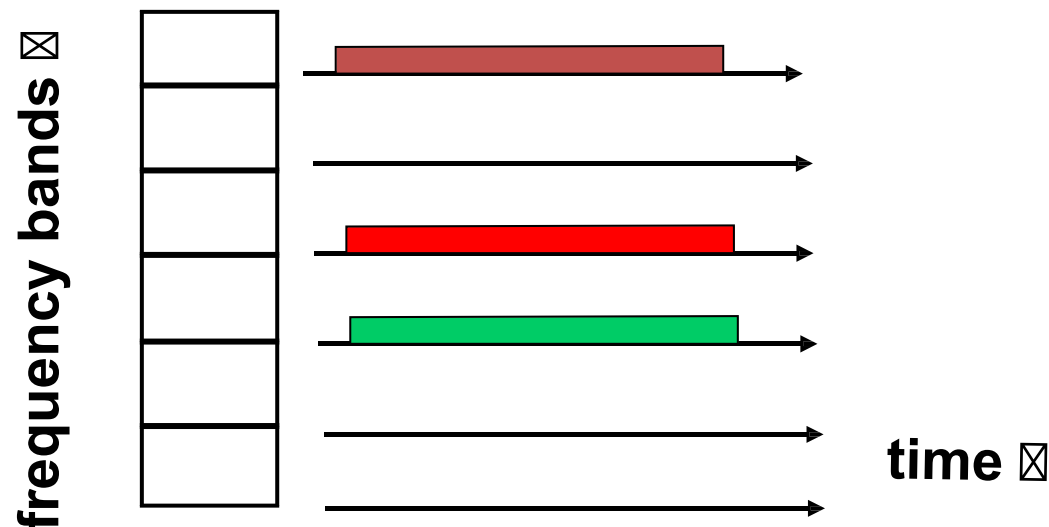
- Channel time is divided into **fixed-period, repeating rounds**
 - Each user gets a fixed-length **slot** (packet time) in each round
 - Send across all frequencies during this slot
 - **Unused slots are wasted**
 - **Out-of-band:** Mechanism for allocating/de-allocating slots
 - **e.g.:** six stations, only 1, 3, and 4 have data to send
- 6-slot**



Time →

FDMA: Frequency Division Multiple Access

- Channel's frequency range (for all time) divided into frequency ***bands***
 - Each user gets a fixed frequency band (**unused frequency slots are wasted**)
- **e.g.:** six stations, only #1, #3, and #4 have data to send



TDMA and FDMA: Considerations

- **Advantages**

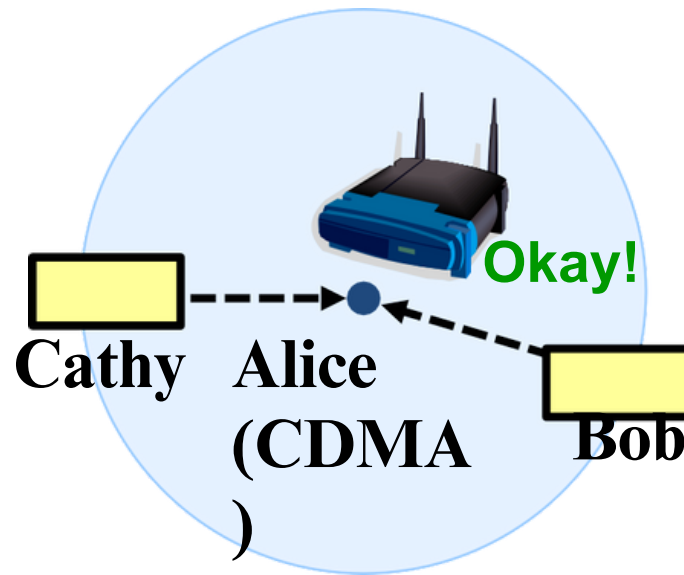
1. Users are **guaranteed** to be able to send bits, **continuously** (FDMA) or **periodically** (TDMA)

- **Disadvantages**

2. Unused time slots or frequency bands **reduce channel utilization**
3. An out-of-band mechanism is needed to allocate slots or bands (which **requires another channel**). Slot assignment in TDMA isn't automatic—it requires coordination over a separate signaling channel.
4. Guard bands or guard times **reduce channel utilization**

CDMA: Code Division Multiple Access

- All users transmit over the **same frequencies**, and **at the same time**:



- Allows multiple users to coexist and transmit simultaneously with **no interference**

Representing bits as binary levels

- Let's represent bits with two (binary) levels as follows:

0 bit $\square\square$ +1 level 1 bit $\square\square$ -1 level

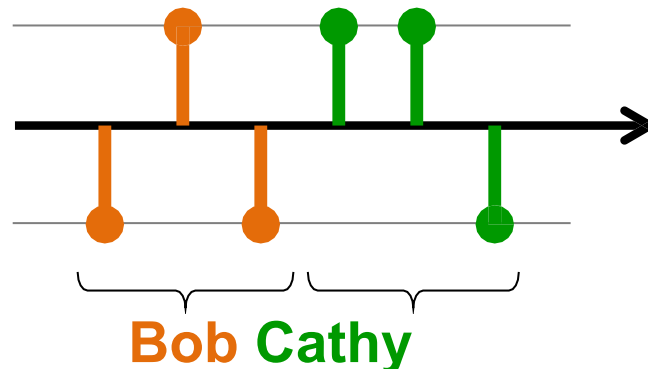
- Scenario:** Alice receives data from **Bob** and **Cathy**:



Cathy Alice Bob

- **TDMA e.g.:** **Bob** sends bits **101**, **Cathy** sends **001**:

+1
time
-1



TDMA timeslots:

CDMA: User codes

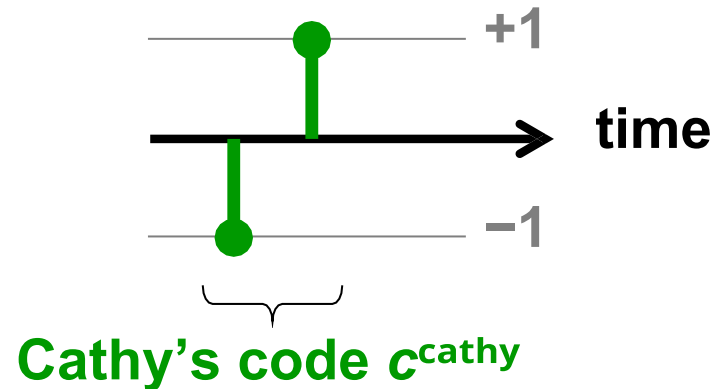
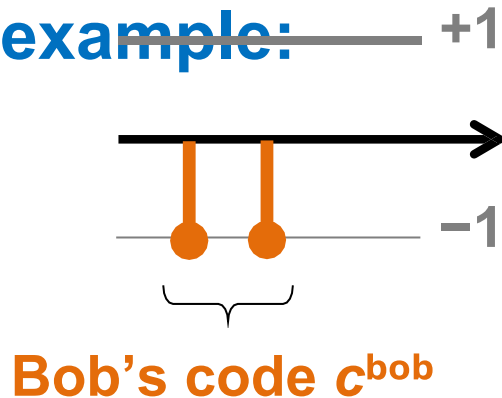


- Assign each user a unique binary sequence of bits:

code

- Call each code bit a **chip** (convention)
- Call the code length **M**

- CDMA example:**

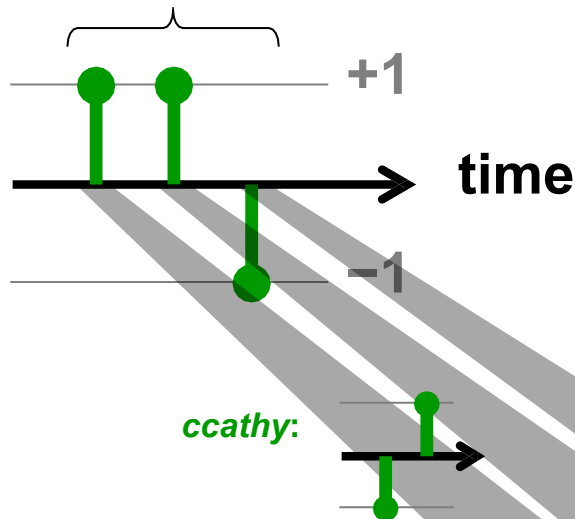


CDMA: Cathy Sending



- Suppose Cathy alone sends message bits

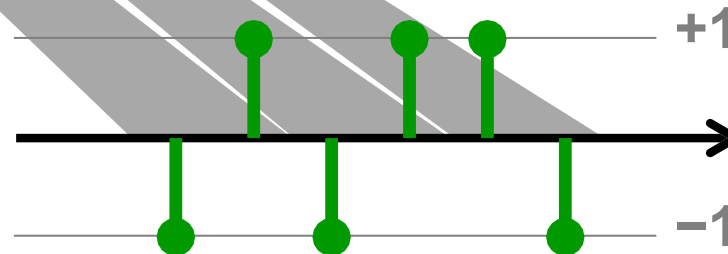
001:
Cathy's message



Algorithm (CDMA encoding):
For each message bit m :
Send $m \times c^{\text{user}}$

L data bits $\boxtimes M \times L$ CDMA chips
Bit rate: Factor of M **slower**

**Cathy's transmitted
CDMA signal:**



CDMA: Assumptions

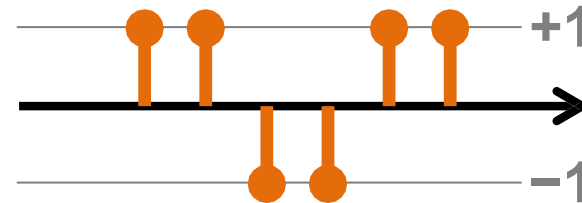


- Let's assume we have a way of:
 - Synchronizing **Cathy's** and **Bob's** data bits in time
 - Synchronizing **Cathy's** and **Bob's** CDMA chips in time
 - Estimating and correcting the **effect of the wireless channel**
between **Cathy** and **Bob** to **Alice**

What Alice Hears



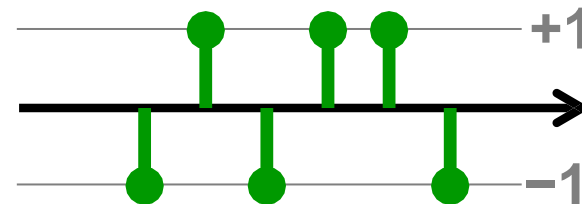
Bob's transmitted
CDMA signal:



+

+

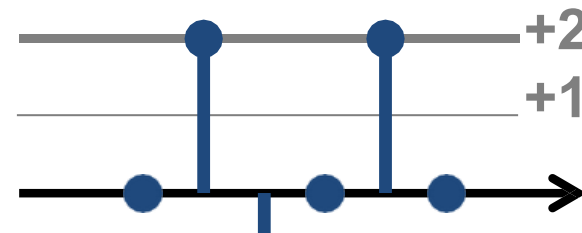
Cathy's transmitted
CDMA signal:



=

=

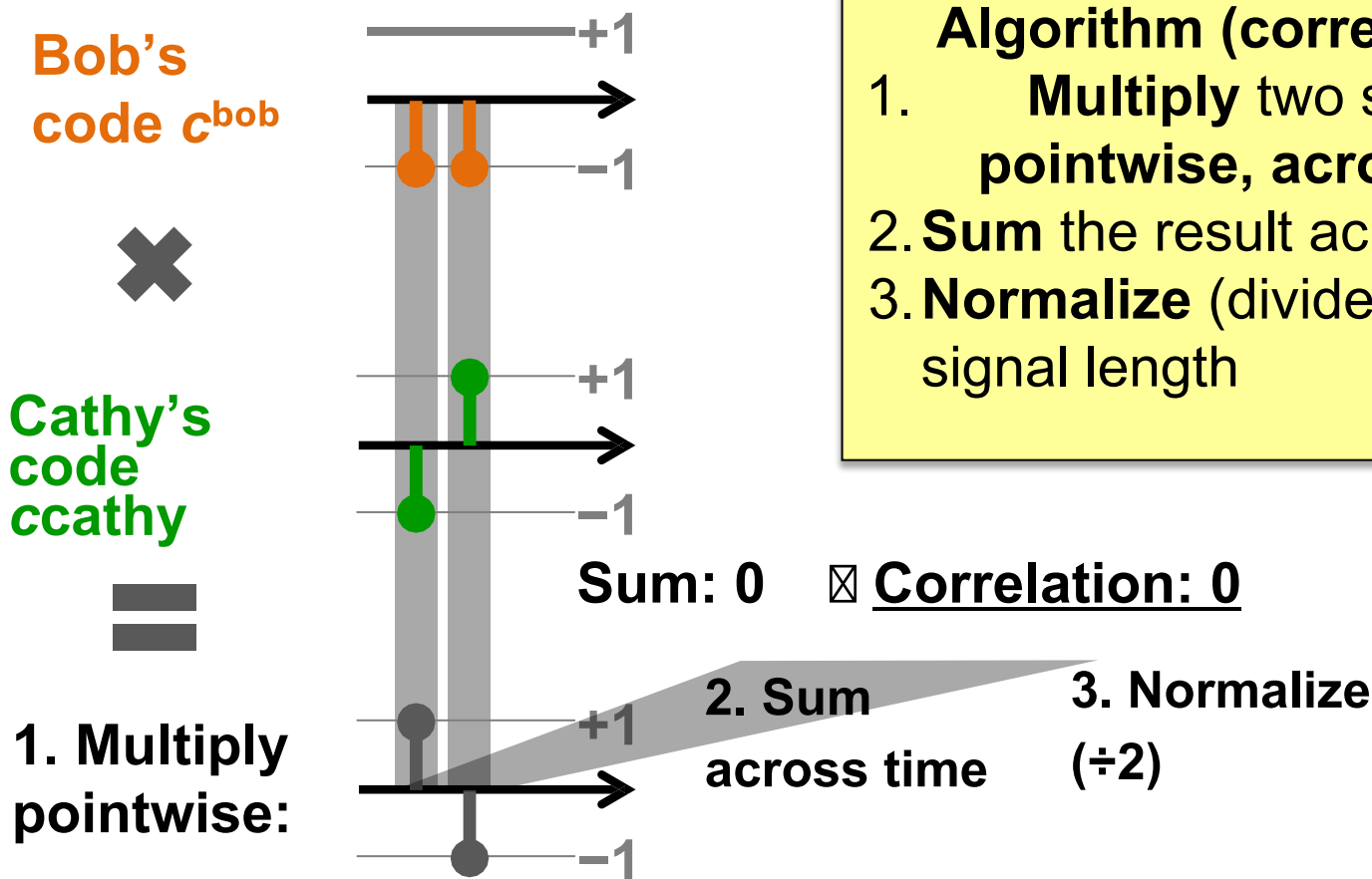
What Alice hears:



Result: Neither Bob nor Cathy's signal – interference!



Tool: Correlation



Algorithm (correlation):

1. **Multiply** two signals pointwise, across time
2. **Sum** the result across time
3. **Normalize** (divide) by the signal length

Tool: Correlation

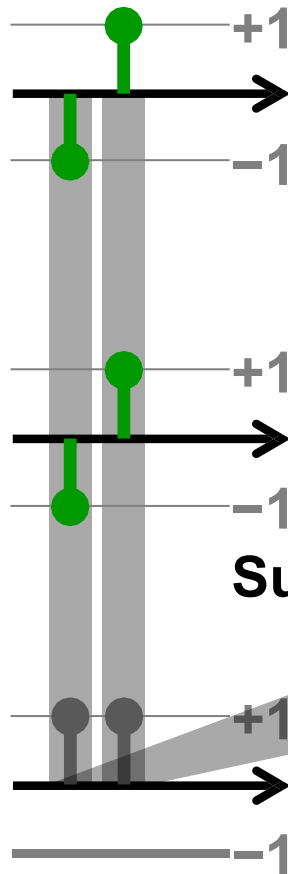
Cathy's
code
ccathy



Cathy's
code
ccathy



1. Multiply
pointwise:



Sum: 2 ✕ Correlation: 1

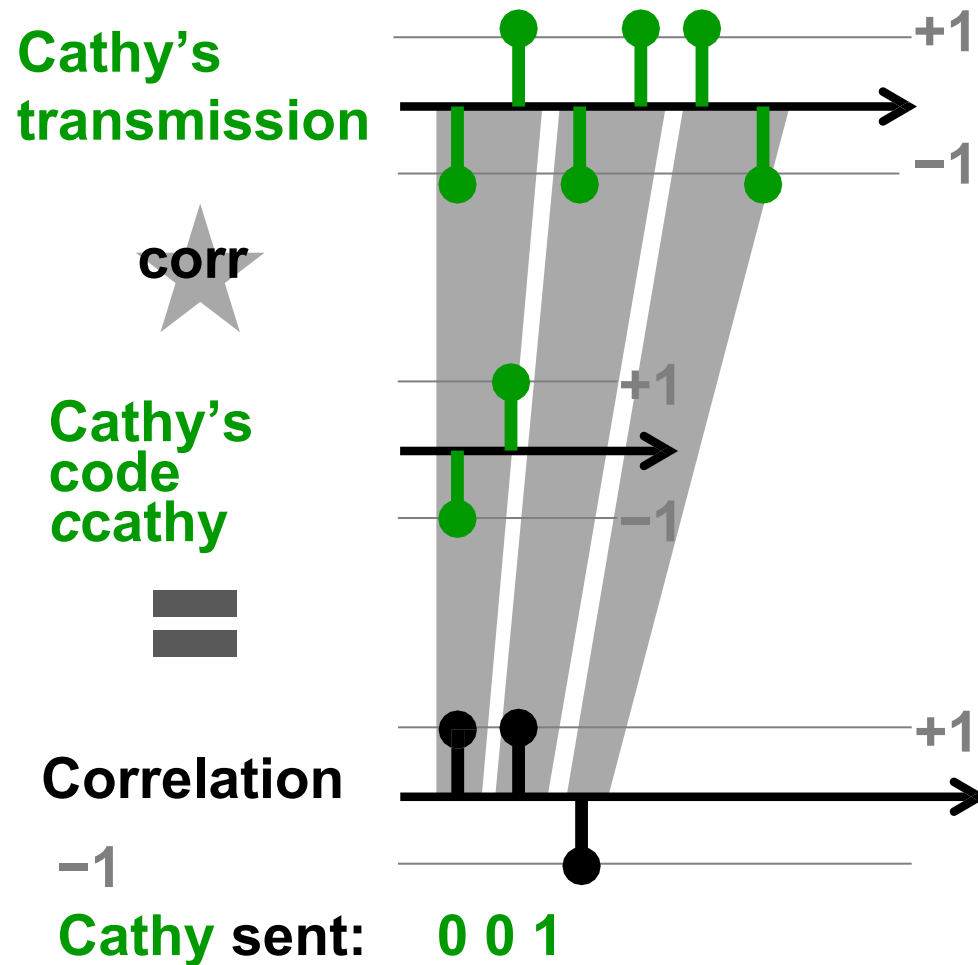
2. Sum
across time

3. Normalize
(÷2)

Algorithm (correlation):

1. **Multiply** two signals
pointwise, across time
2. **Sum** the result across time
3. **Normalize** (divide) by the
signal length

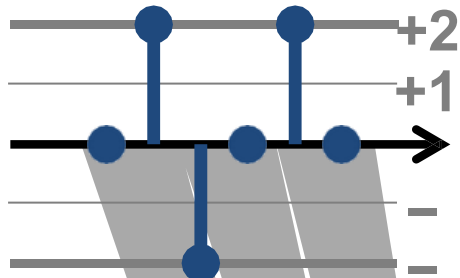
Correlating Cathy's Code with Cathy's CDMA transmission



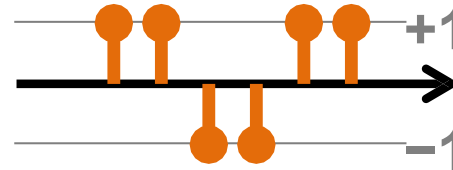
Listening to Cathy



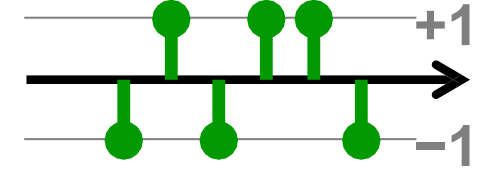
Alice hears
a mixture



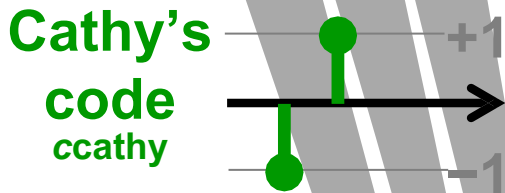
Bob's
transmission



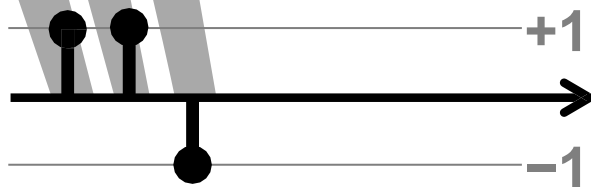
Cathy's
transmission



Zero-correlation with Bob's code **cancels**
Bob's transmission from the mixture



Correlation



Cathy sent: 0 0 1

CDMA: Conclusions

- Communicate **collision-free** over the **same time and same frequency**
- **Tradeoff: Lowered the bit rate**
- *Was it a good tradeoff?*

Today

1. How does channel access differ between wireless and wired networks?
2. Medium access control: Provisioned protocols
- 3. Contention-based medium access control**

Idea: Use a single, shared channel

- Spread transmission across whole 100 MHz of spectrum
 - **Remove constraints** assoc. w/one channel per user
 - No need for strict **one-channel-per-user** allocations.
 - Users can dynamically access any part of the shared channel, making it more efficient and flexible
 - **Robust to multi-path fading**
 - In wireless environments, signals reflect and arrive along multiple paths
 - Spreading the signal across many frequencies ensures that even if some parts are lost, others may arrive intact
 - **Supports peer-to-peer communication**
- **Collisions:** Receiver must hear ≤ 1 strong transmission at a time
- So adopt **deference** from
 - **Listen** (*carrier sense*, **CS**) before sending, **defer** to ongoing

Concurrency versus Taking Turns

- Far-apart links should **send concurrently** (*spatial reuse*)

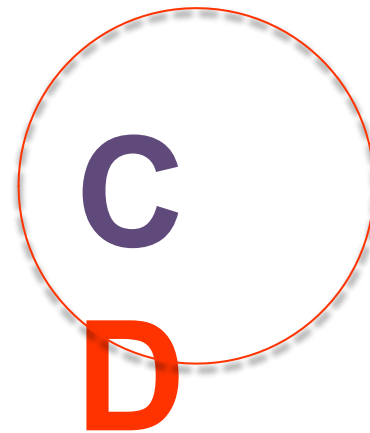
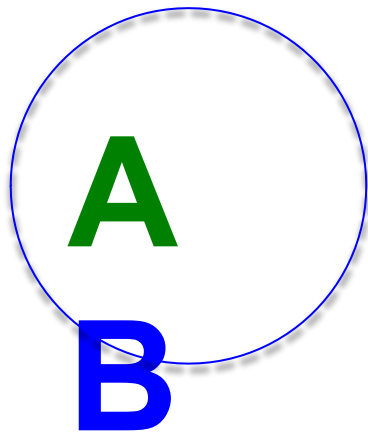


- Nearby links should **take turns**:



When Does CS Work Well?

- Two transmission pairs are **far away** from each other
 - **Neither sender** carrier-senses the other

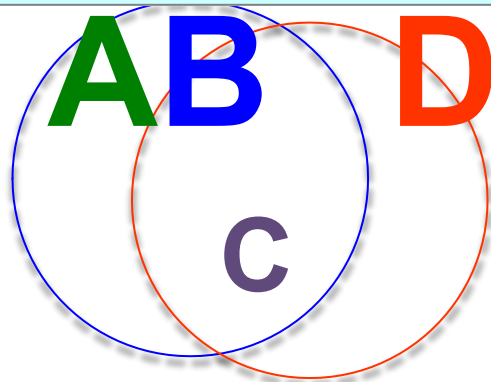


B transmits to A, **while** D transmits to C.

When Does CS Work Well?

- Both transmitters **can carrier sense** each other
 - Carrier sense uses **thresholded correlation** value to determine if medium occupied

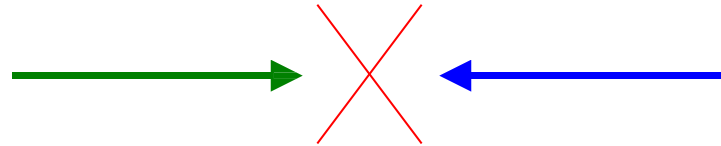
But what about cases in between these extremes?



B transmits to A, D transmits to C, **taking turns.**

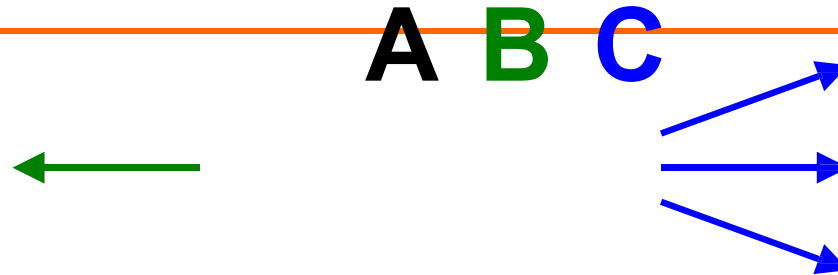
Hidden Terminal Problem

A B C



- **C** can't hear **A**, so **C** will transmit while **A** transmits
 - **Result: Collision at B**
- Carrier Sense insufficient to detect all transmissions on wireless networks!
- **Key insight:** Collisions are spatially located at

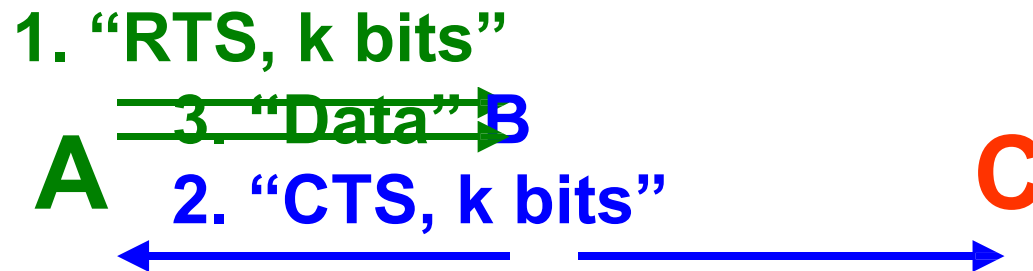
Exposed Terminal Problem



- If C transmits, does it cause a collision at A? **No!**
 - **Yet C cannot transmit while B transmits to A**
- Same insight: **Collisions spatially located at receiver**

RTS/CTS

- Exchange of two short messages: *Request to Send (RTS)* and *Clear to Send (CTS)*
- Algorithm
 1. A sends an **RTS** (tells B to prepare) with message length k
 2. B replies an **CTS** (echoes message length)
 3. A sends its **Data**



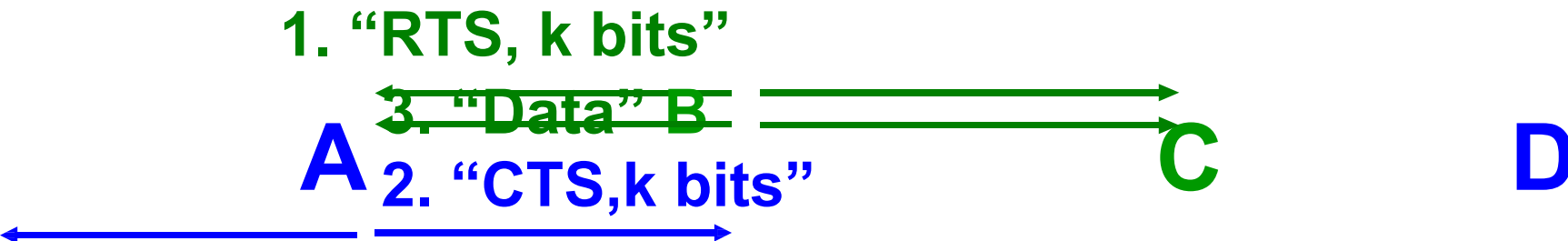
Deference to CTS

- **Hear CTS** ☒ Defer for **length of expected data** transmission time
- **Solves hidden terminal** problem



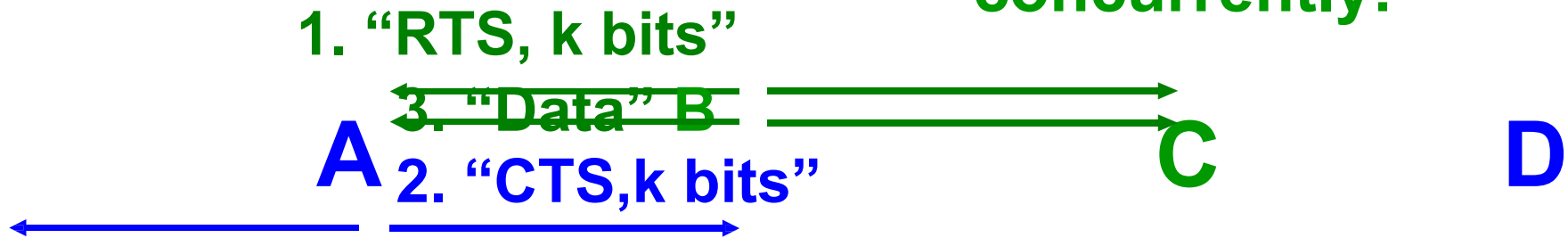
Deference to RTS, but not CS

- Hear -> **RTS** , -> **No CTS** , -> **Data**
 - So terminals **C** **Do not transmit to D !**
 - **But Could transmit to D !**



Deference to RTS, but not CS

So **exposed** terminals **B, C** can transmit concurrently:



✓ Common Solutions:

1. RTS/CTS Mechanism (Request to Send / Clear to Send)

- Node B sends an RTS to A.
- A replies with a CTS.
- Node C hears **RTS from B** but **does not hear CTS from A** (because A is out of C's range).
- ➤ **C knows it won't interfere with A** and can proceed with its own transmission safely.

✓ **Insight:** RTS/CTS provides **receiver-based collision control**, not just sender-based carrier sensing.